Exterior Grille WPL/WPLS



Description
Weatherproof rectangular aluminum ventilation grille, providing protection against ingress of rain, snow, leaves, birds and insects.

Application
Supply/ exhaust of the airflow

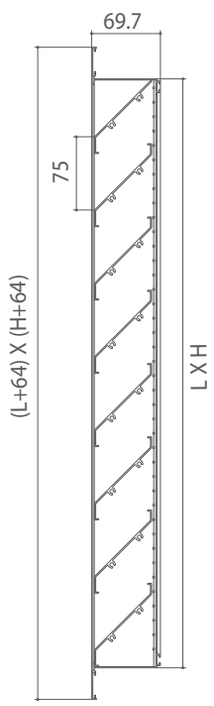
Characteristics
The grille is provided with weatherproof blades, galvanized steel wire mesh and mounting holes for fixing.
Wire mesh used for **WPL** grille is 10 x 10 mm and 6.3 mm x 6.3 mm for **WPLS**
Available sizes:
WPL any size from 800 x 800 mm to 2000 x 2000 mm.
WPLS any size from 100 x 100 mm to 800 x 800 mm.
Modular execution for large dimensions

Accessories
Air filter **FIL**
Insulated/Non-insulated adapter **ADP IZ/N**

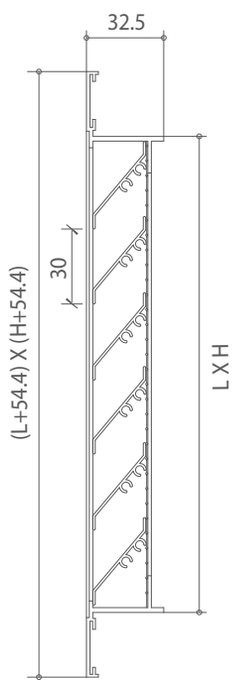
Materials and finishing
Extruded aluminum, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural anodised aluminum.

Technical design

WPL



WPLS



Assembly Method
Standard mounting, with screws.

Selection table WPL

Air flow (m^3/h)	Ak [m^2]	0.30	0.40	0.50	0.60	0.80	1.00	1.5	2.00	2.5	3.00
3000	Veff [m/s] ΔPt [pa]	2.9 10	2.2 12.0								
4000	Veff [m/s] ΔPt [pa]	4 15	2.9 25.0	2.4 14.0							
5000	Veff [m/s] ΔPt [pa]	4.4 25	3.6 40.0	2.9 24.0	2.4 15.0						
6000	Veff [m/s] ΔPt [pa]	5.6 30	4.2 54.0	3.3 35.0	2.9 24.0	2.1 8					
7000	Veff [m/s] ΔPt [pa]	6.5 28	4.9 77.0	4.0 48.0	3.5 37.0	2.4 15.0					
8000	Veff [m/s] ΔPt [pa]			4.6 63.0	3.9 53.0	2.8 24.0	2.3 10				
9000	Veff [m/s] ΔPt [pa]			5.4 90.0	4.5 63.0	3.2 34.0	2.7 20.0				
10000	Veff [m/s] ΔPt [pa]				4.9 77.0	3.5 37.0	2.8 24.0				
20000	Veff [m/s] ΔPt [pa]						5.4 90.0	3.8 45.0	2.8 23.0	2.3 14.0	
30000	Veff [m/s] ΔPt [pa]							5.7 92.0	4.0 53.0	3.3 35.0	2.8 23.0
40000	Veff [m/s] ΔPt [pa]								5.4 90.0	4.5 63.0	3.8 45.0

Discharge area Ak [m^2]									
Height (mm)	Length (mm)								
	800	900	1000	1100	1200	1400	1600	1800	2000
800	0.36	0.40	0.45	0.50	0.55	0.64	0.74	0.83	0.93
900	0.40	0.45	0.51	0.56	0.61	0.72	0.83	0.93	1.04
1000	0.45	0.51	0.57	0.63	0.69	0.81	0.94	1.06	1.18
1100	0.51	0.57	0.64	0.71	0.78	0.91	1.05	1.08	1.32
1200	0.55	0.62	0.70	0.77	0.84	0.99	1.14	1.28	1.43
1400	0.66	0.74	0.83	0.92	1.01	1.18	1.36	1.53	1.71
1600	0.75	0.85	0.95	1.05	1.15	1.36	1.56	1.76	1.96
1800	0.85	0.96	1.08	1.19	1.30	1.53	1.76	1.98	2.21
2000	0.95	1.08	1.21	1.34	1.47	1.72	1.98	2.23	2.49

Selection table WPLS

Air flow (m^3/h)	Ak [m^2]	0.01	0.02	0.03	0.04	0.05	0.06	0.08	0.10	0.15	0.20	0.25	0.30	0.40
200	Veff [m/s] ΔPt [pa]	5.6 50	2.8 8.0											
300	Veff [m/s] ΔPt [pa]		4.2 28.0	2.8 8.0										
400	Veff[m/s] ΔPt [pa]		5.6 50.0	3.9 26.0	2.8 8.0	2.3 3.0								
500	Veff [m/s] ΔPt [pa]			4.8 38.0	3.6 20.0	2.8 8	2.3 3.0							
600	Veff [m/s] ΔPt [pa]			5.8 53.0	4.3 29.0	3.4 18.0	3.0 14.0	2.2 2						
700	Veff [m/s] ΔPt [pa]				5 40	4.0 25.0	3.4 18.0	2.5 7.0	2.2 2					
800	Veff [m/s] ΔPt [pa]				5.8 53.0	4.7 35.0	3.8 25	2.8 8	2.4 4.0					
1000	Veff [m/s] ΔPt [pa]					5.5 50	4.8 38.0	3.5 18.0	2.8 8.0					
2000	Veff [m/s] ΔPt [pa]								5.8 53.0	3.8 25.0	2.8 8.0	2.4 4.0		
3000	Veff [m/s] ΔPt [pa]									5.8 53.0	4.4 30	3.5 19.0	2.8 8.0	2.2 2.0
4000	Veff [m/s] ΔPt [pa]										5.6 51.0	4.6 32.0	3.8 25	2.8 8.0
5000	Veff [m/s] ΔPt [pa]											5.5 50.0	4.8 37.0	3.5 18.0

Discharge area Ak [m^2]								
Height (mm)	Length (mm)							
	150	200	300	400	500	600	700	800
100	0.001	0.008	0.012	0.017	0.024	0.028	0.040	0.050
150	0.013	0.017	0.025	0.030	0.040	0.050	0.060	0.070
200	0.018	0.020	0.030	0.040	0.060	0.070	0.080	0.090
300	0.027	0.035	0.050	0.070	0.090	0.100	0.120	0.140
400	0.036	0.045	0.070	0.090	0.120	0.140	0.170	0.190
500	0.045	0.060	0.090	0.120	0.150	0.180	0.210	0.240
600	0.050	0.070	0.100	0.130	0.180	0.210	0.250	0.290
700	0.060	0.080	0.110	0.160	0.210	0.250	0.300	0.330
800	0.070	0.090	0.130	0.180	0.240	0.300	0.340	0.400

Annotation

* Ak [m^2] - Free surface of the grille
* ΔPt [pa] - Pressure loss
* Veff[m/s] - Air velocity inside the grille

Note: For different discharge areas than those recorded in selection tables, air velocity and pressure drop will be obtained by interpolation.

Order codes

Type + **FIL** + **ADP IZ/N** / **Dimensions** / **L x H** / **O**

Type
- **WPL**
- **WPLS**

FIL
- Air filter

ADP IZ/N
- Insulated/Non-insulated adapter

Dimensions

WPL	WPLS
800 x 800	150 x 150
900 X 900	200 x 200
1000 X 1000	300 x 300
1100 X 1100	400 x 400
1200 X 1200	500 x 500
1400 X 1400	600 x 600
1600 X 1600	700 x 700
1800 X 1800	800 x 800
2000 X 2000	

L x H
- Other dimensions on request

O
- **0** Painted white RAL 9016
- **R** Other RAL paints
- Natural anodised